

Simple sections of polyhedra

Cutting a solid with a plane — the three rules that make the cut, and what shape appears.

LESSON 12–13

2 × 40 MIN

GEOMETRY 10, §7

IGCSE E4.8

LESSON CLOCK

10'

20'

24'

18'

16'

12'

What a section is	10 min
The three construction rules	20 min
Sections of a cube	24 min
Areas of sections	18 min
Practice	16 min
Homework	12 min

LEARNING OBJECTIVES

- Construct the section of a cube or prism through three given points.
- Apply the rule that a plane cuts two parallel faces in parallel lines.
- Name the polygon obtained and justify the number of sides.
- Compute the area of a simple section.

TEXTBOOK

National	pp. 61–70
Cambridge	Core & Extended, pp. 294–298
Workbook	Exercise 7.1

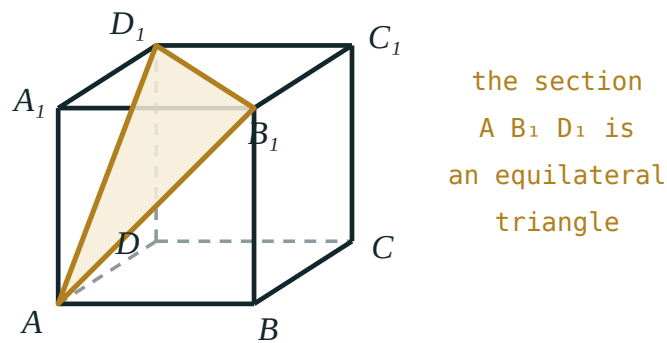
01

Explanation

The three rules

EVERYTHING FOLLOWS FROM THESE

R1 Two points of the cutting plane on the same face join by a straight segment in that face. **R2** A plane cuts two parallel faces in two **parallel** lines. **R3** The section is a closed polygon; every side lies in a face of the solid.



A plane through three points of a cube. R2 forces the top and bottom cuts to be parallel.

R2 does most of the work. Once one side of the section is drawn on a face, the side on the opposite face is fixed in direction — only its position remains to be found.

Sections of a cube

A plane can cut a cube in a triangle, a quadrilateral, a pentagon or a hexagon — but never in a heptagon, because a cube has only six faces and each contributes at most one side.

SECTION	THROUGH	AREA FOR SIDE A
square	parallel to a face	a^2
rectangle	a diagonal plane ABC_1D_1	$a^2\sqrt{2}$
equilateral triangle	three vertices next to one corner	$\frac{\sqrt{3}}{2}a^2$
regular hexagon	midpoints of six edges	$\frac{3\sqrt{3}}{4}a^2$

THE HEXAGON IS THE BEAUTIFUL ONE

Cut a cube through the midpoints of six edges, perpendicular to a space diagonal, and the section is a **regular** hexagon of side $\frac{a\sqrt{2}}{2}$. It is worth building once in card.

Constructing a section, step by step

Problem. Cube $ABCD A_1 B_1 C_1 D_1$. Construct the section through A , C and B_1 .

1. A and C are both on the bottom face: draw AC (R1).
2. A and B_1 are both on face ABB_1A_1 : draw AB_1 (R1).
3. C and B_1 are both on face BCC_1B_1 : draw CB_1 (R1).
4. The polygon closes.

All three segments are face diagonals of length $a\sqrt{2}$, so the section is an **equilateral triangle** of area $\frac{\sqrt{3}}{4} \cdot 2a^2 = \frac{\sqrt{3}}{2} a^2$.

WHEN TWO POINTS ARE NOT ON ONE FACE

Extend an existing edge of the section until it meets the plane of a face, producing a new point you can use. That extended line is the **trace** of the cutting plane on that face.

02 Worked examples

EXAMPLE 1 Cube of side 6. Find the area of the diagonal section ABC_1D_1 .

- 1 $AB = 6$
An edge.
- 2 $BC_1 = 6\sqrt{2}$
A face diagonal.
- 3 The section is a rectangle.
- 4 $6 \times 6\sqrt{2} = 36\sqrt{2} \approx 50.9$

ANSWER

$$36\sqrt{2} \approx 50.9 \text{ square units}$$

EXAMPLE 2 Cube of side a . Find the area of the hexagonal section through six edge midpoints.

- ① Each side joins midpoints of adjacent edges.

Side $\frac{a\sqrt{2}}{2}$.

- ② The hexagon is regular.

$$S = \frac{3\sqrt{3}}{2} s^2$$

- ③ $s^2 = \frac{a^2}{2}$

- ④ $S = \frac{3\sqrt{3}}{2} \cdot \frac{a^2}{2} = \frac{3\sqrt{3}}{4} a^2$

ANSWER

$$\frac{3\sqrt{3}}{4} a^2 \approx 1.30a^2$$

EXAMPLE 3 What is the maximum number of sides a section of a cube can have, and why?

- ① Each side lies in one face.

A cube has six faces.

- ② A plane meets each face in at most one segment.

- ③ Six is attained by the midpoint hexagon.

ANSWER

6

03 Interactive model

Slice a plasticine cube with a wire to show the triangle, the rectangle and the hexagon.

Naming a section

A plane parallel to a face of a cube cuts a:

The section ABC_1D_1 of a cube is a:

The maximum number of sides of a section of a cube:

Two parallel faces are cut by a plane in:

perpendicular lines

parallel lines

a point

nothing

The section through three vertices adjacent to one corner is:

a right triangle

an equilateral triangle

a square

a hexagon

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Section (cross-section)	Kesim	Сечение
Cutting plane	Kesuvchi tekislik	Секущая плоскость
Trace of a plane	Tekislik izi	След плоскости
Diagonal section	Diagonal kesim	Диагональное сечение
Parallel faces	Parallel yoqlar	Параллельные грани
Hexagonal section	Olti burchakli kesim	Шестиугольное сечение
Construction of a section	Kesim yasash	Построение сечения
Midpoint of an edge	Qirra o'rtasi	Середина ребра
Regular hexagon	Muntazam olti burchak	Правильный шестиугольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A section is:

a shadow

the figure where a plane cuts a solid

a net

a projection

Rule R2 says a plane cuts parallel faces in:

equal lines

parallel lines

perpendicular lines

points

The area of the diagonal section of a cube of side a is:

$$a^2$$

$$a^2\sqrt{2}$$

$$2a^2$$

$$a^2\sqrt{3}$$

A cube section can never be a:

triangle

pentagon

hexagon

heptagon

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Section of a cube parallel to a face

ANSWER a square

2. Area of that section, side 5

ANSWER 25

3. Diagonal section of a cube of side 4 — its area

ANSWER $16\sqrt{2}$

4. Maximum sides of a cube section

ANSWER 6

5. Section through A , C , B_1 is

ANSWER an equilateral triangle

6. What does R2 say?

ANSWER Parallel faces give parallel cuts

7. Can a cube section be a regular pentagon?

ANSWER no

Medium · the standard question

7 PROBLEMS

1. Cube side 6; area of the section through A, C, B_1

ANSWER $18\sqrt{3} \approx 31.2$

2. Cube side a ; area of the midpoint hexagon

ANSWER $\frac{3\sqrt{3}}{4}a^2$

3. Cube side 4; area of the midpoint hexagon

ANSWER $12\sqrt{3} \approx 20.8$

4. Section of a triangular prism parallel to the base

ANSWER a congruent triangle

5. Section of a square pyramid parallel to the base

ANSWER a smaller square

6. Cube side 10; diagonal section area

ANSWER $100\sqrt{2} \approx 141.4$

7. Section of a cube through 5 faces is a

ANSWER pentagon

Hard · stretch and reason

7 PROBLEMS

1. Cube $ABCD A_1 B_1 C_1 D_1$ side 6. Construct and find the area of the section through A , midpoint of BB_1 , and C

ANSWER A rhombus of area $18\sqrt{5} \approx 40.2$

2. Prove the midpoint section of a cube is a regular hexagon

ANSWER All six sides equal $\frac{a\sqrt{2}}{2}$ by symmetry about the space diagonal

3. Why is that hexagon perpendicular to a space diagonal?

ANSWER The diagonal is an axis of order 3 through the hexagon centre

4. Cube side a . Find the largest possible area of a plane section

ANSWER $a^2\sqrt{2}$ — the diagonal rectangle

5. Section of a regular tetrahedron through the midpoints of four edges

ANSWER a square

6. Prove that section is a square

ANSWER Each side is a midline, half an opposite edge, and the pairs are perpendicular

7. A plane cuts a cube in a pentagon. How many faces does it miss?

ANSWER 1

07 Homework

Homework — 5 tasks

Constructions must show the rule used at each step (R1, R2 or R3).

1. Construct the section of a cube through A , C and D_1 and name the shape.
2. Cube of side 8: find the area of the diagonal section.
3. Construct the regular hexagonal section and find its area for side 8.
4. Explain why a section of a cube cannot have seven sides.

5. Construct the section of a square pyramid through the midpoints of two base edges and the apex.